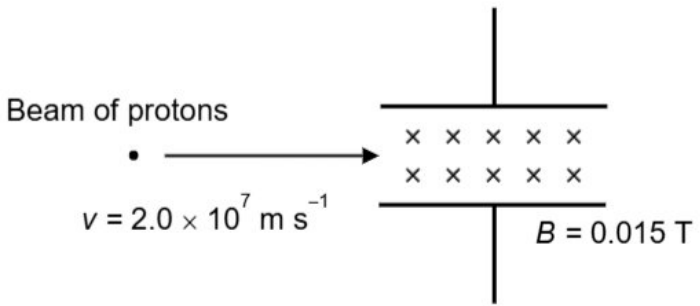
25 A beam of protons passes through a velocity selector as shown in the diagram below. The magnetic flux density, B is 0.015 T and directed into the plane of the paper



If protons travelling at 2.0×10^7 m s^{-1} pass through undeflected, what would be the direction and magnitude of the electric field?

	direction	magnitude
A	downwards	1.3×10^3 N C^{-1}
B	upwards	1.3×10^3 N C^{-1}
C	downwards	3.0×10^5 N C^{-1}
D	upwards	3.0×10^5 N C^{-1}